

1 Conclusion and Future Work

1.a Summary of Thesis Findings

This thesis has systematically investigated the feasibility, theoretical scaling, and physical limitations of Over-the-Air (OTA) computation. By developing a comprehensive cross-platform modeling methodology—spanning mathematical derivation, discrete-time MATLAB simulation, continuous-time Simulink architectures, and Software-Defined Radio (SDR) realization—the research bridged the significant gap between idealized analog computation theory and practical deployment.

The fundamental analysis confirmed that under ideal conditions, Over-the-Air (OTA) computation achieves unprecedented spectral efficiency by exploiting the superposition property of the wireless Multiple-Access Channel. However, the rigorous evaluation of channel impairments demonstrated that the system's performance is severely bottlenecked by uncoordinated transmitter dynamics. Specifically, multipath propagation and fractional timing delays induce Intersymbol Interference (ISI) that manifests as an irreducible error floor. Furthermore, asynchronous Carrier Frequency Offsets (CFO) cause catastrophic coherent demodulation failure due to beat-frequency oscillations. While non-coherent envelope detection provides a robust fallback, it suffers from vector magnitude fading. The findings conclusively show that traditional point-to-point equalization schemes (e.g., standard Costas loops) are insufficient for multi-user aggregated signals, necessitating paradigm shifts in how synchronization and filtering are handled for OTA networks.

1.b Future Research Directions

The insights gained from the physical degradation models open several critical pathways for future research in OTA computation and federated edge learning [2], [3]:

- **Adaptive and Non-Linear Equalization:** Transitioning from static recovery loops to dynamic, adaptive algorithms is paramount. Future work will explore the deployment of Unscented Kalman Filters (UKF) and Extended Kalman Filters (EKF) for the real-time tracking and compensation of non-linear impairments directly from the aggregated waveform.
- **Multi-Antenna and MIMO Systems:** Expanding the single-antenna architecture to Multiple-Input Multiple-Output (MIMO) systems offers spatial degrees of freedom to combat fading. Implementing spatial equalization and receive beamforming [1] could significantly mitigate the destructive interference caused by asynchronous phase offsets.
- **Large-Scale Federated Learning Deployments:** As OTA computation proves to be a natural fit for the aggregation step in Federated Learning, future physical deployments should evaluate the impact of the quantified OTA error floors on the convergence rates of distributed neural network training.

References

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